

Ikram Badaoui

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Final-year Computer Systems Engineering student at ESI Algiers seeking a 9-month research internship, starting September 2026. Passionate about AI and networking, with practical experience in secure network design and datacenter infrastructures.

EDUCATION

Higher National School of Computer Science (ESI)

Since 2022

Algiers

- Relevant Coursework: Cyber Security, Advanced Network, Machine Learning, Software Architecture, High-Performance Computing, IoT, Algorithm Design and Analysis, Combinatorial optimization, Data Analysis.
- Expected Graduation: June 2027

Scientific Baccalaureate

July 2022

Les Frères Drif High School, Boumerdès

Grade: 17.98/20 (Very Good Honors)

TECHNICAL SKILLS

Infrastructure	ContaineLab, GNS3, Cisco IOS, VMware, KVM, Docker
Programming	Python (NumPy, Pandas, Scikit-learn, PyTorch), C, Java, TensorFlow, ROS
Web & Data	React, Next.js, Angular, FastAPI, Oracle DB, SQL, MQTT (IoT)
Languages	Arabic (native), French (fluent), English (fluent), Turkish (good), Spanish (A1)

PROJECTS

Modern Data Center for ESI: Architecture & Deployment

Feb 2026 – Present

- Owned the implementation of a centralized AAA infrastructure: TACACS+ for administrative device management, RADIUS for network access control, and OpenLDAP for unified RBAC.
- Designed the QoS (Quality of Service) and traffic shaping policies.
- Implemented the redundancy using BGP EVPN and ESI Multihoming (Ethernet Segment Identifier).
- Collaborated on the Spine-Leaf fabric featuring dual-ISP BGP multihoming for redundant Internet access, HA firewalls, and an Ansible-automated deployment for the observability stack, and many services (FTP, DHCP, DNS,...).

ESIBot: Integrated Autonomous Mobile Robot -Team Leader

Feb 2026 – Present

- Led a team of 7 to build an autonomous ROS2 Humble robot, featuring a custom 3D-printed chassis and a distributed ESP32-CAM / Raspberry Pi 4 architecture.
- Engineered a 360° "Pseudo-LiDAR" using 4 ultrasonic sensors on a servo-motor.
- Integrated IMU-based odometry and L298N motor control to achieve precise autonomous navigation and obstacle avoidance.
- Developed a web dashboard integrated with robot for real-time SLAM visualization, video streaming, battery state, and remote teleoperation.

Adaptive Metaheuristic Solver for Bin Packing Problem using Reinforcement Learning

May 2026

- Implemented a Genetic Algorithm for the Bin Packing Problem using population-based search, crossover, mutation, and local improvement.
- Developed a Reinforcement Learning-guided optimizer to adapt search decisions and improve packing quality over benchmark instances.
- Built an ML-guided Hybrid Cuckoo Search Genetic Algorithm that selects adaptive search profiles and records solution quality, runtime, and gap metrics.

Cybersecurity — Practical Labs

Feb 2026 - Present

- Applied cryptographic protocols and encryption tools (OpenSSL, CrypTool) for secure communication analysis.

- Performed offensive security techniques including password cracking, credential dumping (Mimikatz), and active network sniffing.
- Conducted session hijacking attacks using modern HTTP interception proxies (Caido, Hetty).

EXPERIENCE

- Sonatrach (Drilling Division)**

Aug 2025 – Sept 2025

Network & Security Engineering Intern

 - Designed and simulated a Data Center architecture in GNS3, implementing VLAN/DMZ segmentation and secure VPN tunnels for critical service isolation.
 - Conducted security stress-testing by simulating multiple attacks to validate the effectiveness of implemented DHCP and HSRP/GLBP redundancy, identify vulnerabilities, then implement appropriate corrective actions.
- Training Camp XII Hackathon (2nd Place, Agentic AI Category)**

2026

 - Engineered an automated enterprise ticketing pipeline with multilingual support, developing the React interface and orchestrator integration for multi-agent OCR, FAISS RAG, and Mistral-LLM automated response logic.
- Scientific Carbon Footprint Calculator**

2023 – 2024

 - Developed a React-based analytical tool for a University of Boumerdès researcher to assess product environmental impact and carbon footprint using a database.

CERTIFICATIONS & TRAINING

- Cisco CCNA 1, 2 & 3 — Routing, Switching, Security & Automation

Cisco Networking Academy
- Backend Development with FastAPI & PostgreSQL

Training Camp XI Hackathon
- Agentic AI — Autonomous AI Agents Design, Orchestration & Deployment

Training Camp XII Hackathon

INTERESTS & EXTRACURRICULAR

- **Motion Designer & Mentor — ETIC Club:** Produced motion design and visual storytelling content using Adobe After Effects while mentoring club members in animation fundamentals.
- **Design:** Passionate about graphic design.
- **Reading:** Novels and articles.
- **Traveling:** Nature and cultural exploration.